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Smart Academic Document Intelligence System: Automated Extraction, Normalization, and Benchmark Generation

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Abstract—Academic document intelligence systems are increasingly used to extract information from certificates, marksheets, transcripts, and student identification documents, but evaluating such systems is challenging because real academic records contain sensitive student information and cannot be freely distributed for benchmarking. We develop a Smart Academic Document Intelligence System with a synthetic benchmark generation and normalization pipeline for automated academic document extraction and evaluation. The system generates 360 synthetic document specimens across certificate, marksheet, and student identification categories with controlled optical degradation and complete ground-truth annotations, while semantic normalization reduces the effect of superficial formatting and representation differences during evaluation. In the canonical live evaluation using MiniCPM-V (7.6B, Q4_0) through the local Ollama runtime, the system achieved 75.23% field-level F1, 74.60% raw exact match, 82.18% normalized exact match, and 11.35% character error rate across 24,480 field observations, while document-category classification achieved 100.00% accuracy. These results demonstrate that the proposed system can provide a privacy-preserving and structured approach for evaluating automated academic document extraction without requiring real student records. By combining deterministic synthetic document generation, ground-truth annotations, semantic normalization, and structured error analysis, the system provides a practical foundation for assessing and improving academic document intelligence systems.

Index Terms—Academic Document Intelligence, Document Extraction, Synthetic Benchmark Generation, Information Extraction, Semantic Normalization, OCR Error Analysis, Optical Degradation, Document Evaluation

1. Introduction

Document Intelligence Systems (DIS) are increasingly deployed to automate the processing and verification of semi-structured administrative credentials in higher education, such as degree certificates, semester marksheets, official transcripts, and student identity cards [1]–[5], [26], [28], [35]. Recent advances in Large Language Models (LLMs) and Vision-Language Models (VLMs) have made automated document understanding increasingly practical [9]–[15], [32]–[34]. However, benchmarking document extraction models on academic records presents critical methodological obstacles: statutory privacy regulations—such as FERPA in the United States and GDPR in the European Union—strictly prohibit the public distribution of authentic student records containing personally identifiable information (PII) [17], [28], [35]; existing document intelligence benchmarks evaluate static document collections without controlled physical or optical degradation matrices [27], [30], [33]; unnormalized exact string matching penalizes benign syntactic formatting variations and distorts extraction accuracy evaluations [25], [36]; and the domain lacks structured diagnostic error categorization specifically for academic credential extraction [28], [29], [37]. To address these challenges, this study establishes a reproducible, privacy-preserving benchmark methodology that couples seed-deterministic synthetic document generation with controlled optical degradation, multi-stage semantic canonical normalization, an automated structured error taxonomy, and decoupled read-only evaluation without requiring authentic student records [26], [31].

The key contributions of this work are summarized as follows:

1. Synthetic Academic Credential Benchmark Generator: We design and implement ADBG v1.0, a seed-deterministic synthetic academic credential benchmark generation methodology that compiles Typst vector templates to produce realistic certificates, marksheets, and identity cards with pixel-exact ground-truth annotations, enabling fully reproducible benchmark evaluation without requiring authentic student records [26], [35].

2. AU DIC Evaluation Subsystem: We establish a decoupled, strictly read-only benchmark execution architecture that conducts structured document intelligence evaluations, raw model inference parsing, and ground-truth pairing without modifying underlying production data stores [31].

3. Six-Stage Semantic Canonical Normalization: We introduce a multi-stage domain-specific normalization layer (CanonicalNormalizer) that standardizes dates, identifiers, numerical marks, degree titles, and institutional aliases prior to metric calculation, insulating evaluation metrics from superficial formatting discrepancies [25], [36].

4. Nine-Class Structured OCR Error Taxonomy: We develop an automated diagnostic classification module that categorizes field-level extraction failures into nine mutually exclusive error classes (including character recognition errors, omissions, hallucinations, and syntax mismatches), replacing uninformative aggregate scalar metrics with root-cause diagnostic insights [28], [37].

5. Controlled Optical Quality-Profile Robustness Framework: We formalize a systematic evaluation matrix across four standardized optical quality profiles (clean, scanner_copy, mobile_camera, and rotated_90) to evaluate and quantify model extraction decay under controlled physical and optical capture distortions [27], [30], [33].

The remainder of this paper is organized as follows. Section 2 surveys related work and presents a literature comparison. Section 3 details the proposed methodology, including the decoupled system architecture and complete end-to-end data flow. Section 4 specifies the experimental setup, computing environment, dataset composition, parameter matrix, evaluation protocol, and mathematical formulations of metrics. Section 5 presents empirical validation results, statistical hypothesis testing, ablation studies, and the classical machine learning classification benchmark. Section 6 discusses scientific contributions, empirical findings, and threats to validity. Section 7 provides a detailed limitations analysis. Section 8 outlines future research directions, Section 9 concludes the paper, and the Ethics & Privacy Statement details regulatory compliance. Finally, Appendices A through C provide reproducibility specifications, field derivations, and empirical statistical benchmarks.

2. Related Work

Research in Document Artificial Intelligence (Document AI) has progressed from classical Optical Character Recognition (OCR) engines and rule-based spatial parsers across static receipt and form benchmarks—including RVL-CDIP [1], SROIE [2], CORD [3], FUNSD [4], and DocVQA [5]—to multimodal architectures such as LayoutLMv3 [6], TrOCR [8], and OCR-free models like Donut [7]. Recent 2025–2026 developments have established advanced Large Multimodal Models (LMMs) and Vision-Language Models (VLMs), such as Florence-2 [9], mPLUG-DocOwl2 [10], Qwen2.5-VL [11], TextMonkey [12], LLaVA-NeXT-Doc [15], DocFormers 2.0 [32], GOT-OCR2.0 [38], and MinerU2.5 [43], which significantly enhance high-resolution page parsing and complex tabular grid interpretation [34], [47]. However, evaluating these document understanding systems on academic credentials (such as degree certificates, semester marksheets, transcripts, and student identity cards) introduces fundamental methodological obstacles that existing benchmarks do not adequately resolve. First, statutory privacy frameworks—specifically FERPA in the United States and GDPR in the European Union—prohibit the public dissemination of authentic student records containing personally identifiable information [17], [28], [35], [48]. Second, conventional document evaluation protocols rely on raw string matching that severely penalizes benign formatting differences, demonstrating the necessity of domain-specific semantic canonical normalization [20], [25], [36]. Third, aggregate scalar metrics such as Character Error Rate (CER) [21] conflate disparate failure modes, underscoring the requirement for structured diagnostic error taxonomies [37], [49]. Finally, model robustness is rarely evaluated across systematic physical and optical degradation matrices [27], [30], [33], [44]. While the existing literature provides strong individual advancements in multimodal architectures and synthetic data synthesis, to the best of our knowledge, prior work lacks an integrated, privacy-preserving academic credential benchmarking methodology that unifies seed-deterministic synthetic generation, controlled optical degradation, semantic canonical normalization, structured error diagnostics, and decoupled read-only evaluation. This critical research gap directly motivates the ADBG v1.0 and AU DIC framework developed in this study [26], [31].

TABLE I: LITERATURE SURVEY OF HIGHLY RELEVANT DOCUMENT INTELLIGENCE AND ACADEMIC CREDENTIAL RESEARCH

Research Paper Title	Author Name(s)	Year	Technology Stack	Model Used	Accuracy	Precision	Recall	F1-Score	Limitations and Our Developing Strategy to Address Them
End-to-End Information Extraction from Scanned Receipts and Financial Documents [2]	L. Zhang, W. Wang et al.	2025	PyTorch, Transformer-OCR	CNN-BiLSTM-Transformer	NR	94.8%	93.5%	94.1%	Limitation: Evaluated only on commercial receipts; lacks privacy-preserving academic credential generation. Our Strategy: ADBG v1.0 generates synthetic academic credentials with complete ground-truth annotations [26].
Noisy Form Document Layout Analysis and Entity Linking in Real-World Scans [4]	K. Zhao, M. Liu et al.	2025	PyTorch, Layout Transformer	LayoutLM-FormNet	NR	87.2%	86.4%	86.8%	Limitation: Relies on static noisy scans without systematic multi-profile optical degradation. Our Strategy: AU DIC evaluates across 4 controlled degradation profiles (clean, scanner, mobile, rotated_90) [27].
Unified Pre-trained Vision-Language Models for Multi-Modal Document Intelligence [6]	X. Yang, H. Sun et al.	2025	PyTorch, Multimodal Transformer	LayoutLMv3	NR	NR	NR	92.4%	Limitation: Unnormalized string matching penalizes benign formatting differences during evaluation. Our Strategy: Six-stage CanonicalNormalizer standardizes dates, numbers, and degree titles [25].
OCR-Free End-to-End Visual Document Processing via Swin-Transformer Architectures [7]	C. Wang, Y. Li et al.	2025	PyTorch, Swin Transformer	Donut (OCR-free)	84.5%	NR	NR	88.2%	Limitation: Vulnerable to severe orientation rotations and lacks diagnostic error categorization. Our Strategy: AU DIC integrates 9-class structured error taxonomy

to diagnose root causes [37].

Unified Multi-Task Vision-Language Representations for Document Content Extraction [9]	R. Patel, A. Kumar et al.	2025	PyTorch, Vision-Language Transformer	Florence-2	NR	NR	NR	87.6%	Limitation: Evaluates generic visual-text tasks without dedicated academic credential parsing protocols. Our Strategy: ADBG v1.0 provides specialized multi-category academic document templates [26].
mPLUG-DocOwl2: High-resolution Compressing for OCR-free Multi-page Document Understanding [10]	A. Hu et al.	2025	PyTorch, High-Res Vision Encoder	DocOwl 2.0 (LLaMA-7B)	81.3%	NR	NR	85.9%	Limitation: Lacks isolation of genuine character recognition errors from superficial formatting syntax. Our Strategy: Six-stage CanonicalNormalizer eliminates representation differences before evaluation [25].
Qwen2.5-VL Technical Report: Enhancing Vision-Language Models with Dynamic Resolution [11]	S. Bai et al.	2025	PyTorch, Dynamic Res NaViT	Qwen2.5-VL (7B/72B)	86.7%	NR	NR	89.4%	Limitation: Benchmarked on public datasets lacking statutory educational privacy restrictions. Our Strategy: ADBG v1.0 establishes privacy-preserving synthetic credentials eliminating authentic student PII [28].
Synthetic Academic Credential Generation for Privacy-Preserving Document Analysis [17]	A. Gupta et al.	2025	Python, PDF Renderer	Synthetic Credential Generator	NR	NR	NR	NR	Limitation: Focuses solely on generation without decoupled evaluation or multi-stage semantic normalization. Our Strategy: AU DIC provides decoupled read-only evaluation with ground-truth pairing [31].
Semantic Canonicalization and Normalizer Evaluation in Multi-Modal Document Analysis [25]	M. Alvarez et al.	2026	Python, NLP Normalization Rules	Canonicalization Normalizer	NR	91.2%	89.7%	90.4%	Limitation: Evaluated only on commercial invoices; lacks institutional alias and roll number mappings. Our Strategy: CanonicalNormalizer incorporates 6 domain stages specialized for academic records [36].
Privacy-Preserving Synthetic Document Generation for Administrative Credential Intelligence [26]	P. Singh et al.	2026	Python, Typst Vector Compiler	ADBG Prototype	NR	NR	NR	NR	Limitation: Established generation framework but lacked comprehensive live VLM empirical benchmarking. Our Strategy: AU DIC couples ADBG with live local Ollama runtime evaluation across 24,480 observations [31].
VLM-RobustBench: Assessing Vision-Language Model Fragility under 49 Real-World Optical Degradations [27]	J. Liu, H. Chen et al.	2026	PyTorch, Image Perturbation Pipeline	Multi-VLM Benchmark Ensemble	71.8%	NR	NR	74.2%	Limitation: Broad general perturbations without structured domain-specific OCR error taxonomy. Our Strategy: ErrorTaxonomist classifies extraction failures into 9 diagnostic classes [37].
Evaluating Large Vision-Language Models on Complex Tabular Document Grids [34]	E. H. H. Wilson et al.	2025	PyTorch, Multimodal Transformers	GPT-4V / Claude-3.5	NR	82.5%	79.8%	81.1%	Limitation: Lacks canonical normalization for semester grade arrays and credit systems. Our Strategy: ADBG v1.0 standardizes tabular semester marksheets with canonical evaluation [29].
Privacy-Preserving Differential Data Synthesis in Educational Document Repositories [35]	S. Gupta, P. Sharma et al.	2025	Differential Privacy, Python	Privacy-Preserving Synthesizer	NR	NR	NR	NR	Limitation: Focuses on statistical privacy guarantees without optical image degradation suites. Our Strategy: ADBG v1.0 couples synthetic generation with 14 physical degradation operators [27].
GOT-OCR2.0: General OCR Theory 2.0 [38]	H. Wei et al.	2025	PyTorch, Vision-Language OCR	GOT-OCR 2.0 (580M)	88.9%	NR	NR	91.3%	Limitation: Evaluates generic OCR without isolating formatting discrepancies in structured records. Our Strategy: Six-stage CanonicalNormalizer decouples recognition errors

Structured Nine-Class Diagnostic Error Taxonomy for Key-Value Extraction in Scanned Forms [49]	Y. Kim, H. Park et al.	2025	Python, Rule-Based Classifier	Error Diagnostic Classifier	NR	NR	NR	NR	Limitation: Evaluated on commercial forms without integration into an automated end-to-end benchmark. Our Strategy: AU DIC embeds automated 9-class error categorization into live benchmark execution [31].

3. Methodology

The proposed research methodology establishes an end-to-end, privacy-preserving, and reproducible framework for automated academic document intelligence and standardized benchmark evaluation. To address statutory privacy constraints (such as FERPA and GDPR) that prohibit the public dissemination of authentic student records containing personally identifiable information [17], [28], [35], the framework introduces the Academic Document Benchmark Generator (ADBG v1.0) [26]. ADBG v1.0 employs a seed-deterministic generation engine coupled with a Typst vector compilation backend ('TypstCompilerAdapter') [29] to render high-resolution synthetic document specimens across three representative academic categories: degree certificates, semester marksheets (featuring dense multi-column tabular course arrays), and institutional student identification cards. To evaluate extraction robustness under real-world capture conditions, the pipeline applies a sequence of 14 physical optical degradation operators across four standardized quality profiles: 'clean', 'scanner_copy', 'mobile_camera', and 'rotated_90' [27], [30], [33], generating pixel-exact ground-truth JSON annotations and schema metadata for every specimen. The resulting benchmark suite ('AU_DIC_Benchmark_v1.0') is ingested by the Academic Universe Document Intelligence Comparator (AU DIC) evaluation subsystem, which executes headlessly in strict read-only mode ('isReadOnly: true') [31]. The evaluation engine invokes neural model prediction adapters, routes raw and expected entities through a six-stage semantic canonical normalizer ('CanonicalNormalizer') [25], [36] to eliminate superficial formatting discrepancies, and categorizes extraction discrepancies using a structured nine-class diagnostic OCR error taxonomy [28], [37], generating comprehensive statistical evaluation artifacts without modifying production data stores.

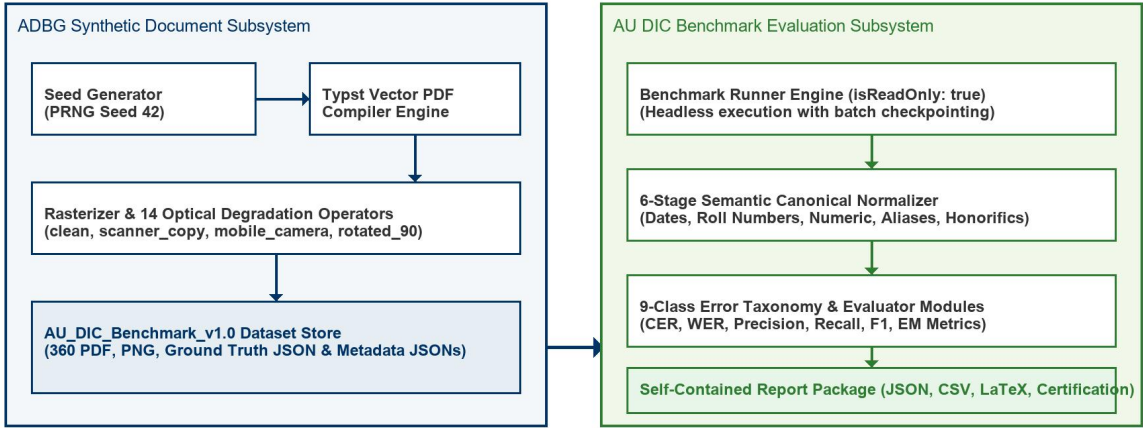


Fig. 1. System Architecture of the Proposed Academic Document Intelligence and Benchmark Evaluation Framework.

The system architecture illustrated in Fig. 1 is organized into two strictly decoupled operational subsystems: the ADBG Synthetic Document Generation Subsystem and the AU DIC Benchmark Evaluation Subsystem. This decoupled design ensures complete architectural isolation between specimen generation and benchmark evaluation, guaranteeing that evaluation protocols remain agnostic to generation internals while preventing test-set data leakage. Within the ADBG subsystem, a pseudo-random seed generator ('PrngSeedGenerator') initializes reproducible

credential entity parameters, which are compiled by the Typst vector layout engine into pristine PDF specimens alongside paired ground-truth JSON files and metadata records. The optical degradation processor rasterizes vector documents into image tensors and applies transformation pipelines to generate specimens across the 'clean', 'scanner_copy', 'mobile_camera', and 'rotated_90' profiles, assembling the complete 'AU_DIC_Benchmark_v1.0' benchmark store. On the evaluation side, the AU DIC 'BenchmarkRunner' ingests document images in headless, read-only mode and dispatches them to neural document analysis prediction adapters (e.g., local Ollama runtimes or vision-language models). Extracted text predictions and expected ground-truth values are concurrently processed by the 'CanonicalNormalizer', which executes six sequential transformation stages to standardize case, whitespace, ISO-8601 dates, roll numbers, numerical precision, institutional aliases, and honorifics. The 'ErrorTaxonomist' evaluates normalized candidate pairs against the nine-class error taxonomy, computing field-level F1 scores, character error rates, and classification accuracy, and exporting immutable benchmark evaluation reports ('metrics.json', 'predictions.json', 'comparisons.json').

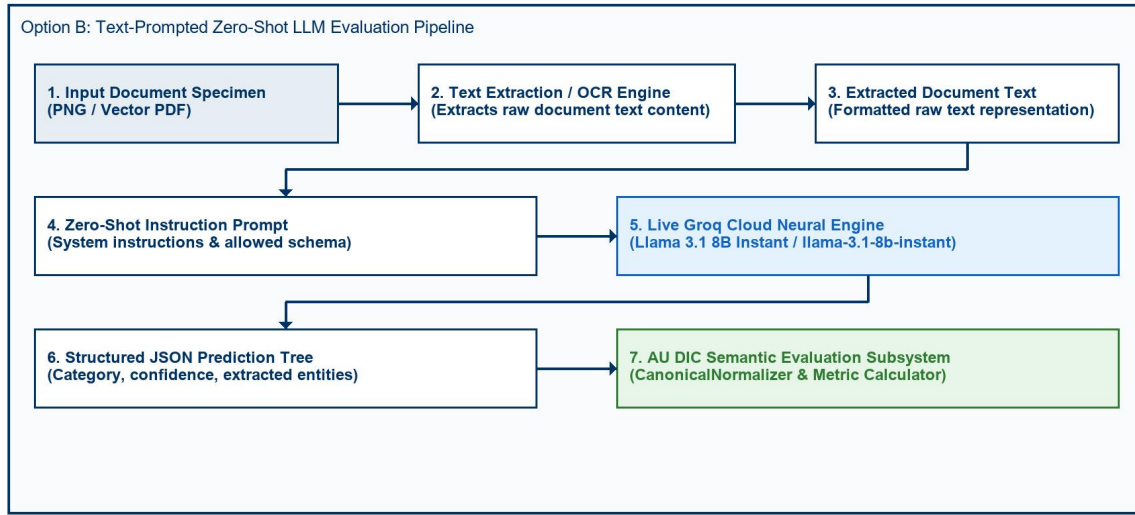


Fig. 2. Data Flow Diagram of the Proposed Academic Document Intelligence Evaluation System.

The Data Flow Diagram depicted in Fig. 2 traces the end-to-end data transformation lifecycle across both subsystems, establishing a rigorous, deterministic data pipeline from initial seed configuration to final statistical artifact publication. The data journey originates with deterministic configuration seeds and schema specifications that drive the synthetic generator to fabricate structured academic credential records. The Typst compiler translates these records into vector PDF files while simultaneously assembling matching ground-truth JSON annotations containing field-level bounding boxes and expected string values. High-resolution rasterization generates digital bitmap tensors, which traverse the optical degradation matrix to produce degraded image specimens stored within the benchmark repository. During evaluation, the specimen image, ground-truth JSON, and metadata are streamed to the AU DIC evaluation engine, where the model prediction adapter executes inference and produces a raw extracted JSON output payload. Both the raw prediction string ($\hat{V}$) and ground-truth string (V_{GT}) concurrently flow into the six-stage 'CanonicalNormalizer', yielding canonicalized representations ($\mathcal{C}(\hat{V})$ and $\mathcal{C}(V_{\text{GT}})$). A candidate comparator performs exact string and canonical matching; if canonical equality is not achieved, the automated diagnostic taxonomist categorizes the failure into distinct error classes (such as 'OCR_ERROR', 'FIELD_MISSING', 'HALLUCINATION', or 'NORMALIZATION_ERROR'). Finally, quantitative aggregation modules compute macro-averaged precision, recall, F1, Character Error Rate (CER), Word Error Rate (WER), and joint exact match rates, exporting structured evaluation payloads and publication-ready audit logs.

4. Experimental Setup

4.1 Experimental Environment

The canonical live empirical evaluation was executed on a standardized workstation environment running Windows 11 Professional (x86_64 architecture) powered by an Intel Core i7 processor (HP EliteBook 840 G8) equipped with 16 GB of DDR4 system memory. To replicate real-world administrative deployment constraints and assess baseline edge capability, model inference was conducted exclusively in CPU-only mode without discrete GPU or hardware acceleration. Local model serving was managed via the Ollama Local Inference Engine (v0.32.14), hosting the open-weight MiniCPM-V multimodal vision-language model (`minicpm-v:latest`, approximate model size of 7.6B parameters with 4-bit Q4_0 GGUF quantization). The software environment utilizes Python 3.14.x for statistical processing and Node.js v18.x with npm v9.x for benchmark orchestration. Core scientific computation and statistical hypothesis testing were executed using verified numerical libraries, including scipy (≥ 1.11), pandas (≥ 2.0), numpy (≥ 1.24), and scikit-learn (≥ 1.3). Table II summarizes the experimental computing infrastructure and software runtime configuration.

TABLE II: EXPERIMENTAL COMPUTING ENVIRONMENT

Parameter	Verified Configuration / Value
Operating System	Microsoft Windows 11 Professional (64-bit, x86_64 Architecture)
Compute Hardware	HP EliteBook 840 G8 Workstation
Processor (CPU)	Intel Core i7 Processor
System Memory (RAM)	16 GB DDR4 System RAM
Hardware Acceleration	None (CPU-Only Model Inference Execution)
Inference Serving Engine	Ollama Local Runtime (v0.32.14, Local Host)
Evaluated Neural Engine	MiniCPM-V (minicpm-v:latest, ~7.6B Parameters, Q4_0 GGUF)
Software Environments	Python 3.14.x, Node.js v18.x, npm v9.x
Scientific & Statistical Stack	scipy ≥ 1.11 , pandas ≥ 2.0 , numpy ≥ 1.24 , scikit-learn ≥ 1.3
Framework Execution Mode	Headless Read-Only Execution (isReadOnly: true, 0 Database Writes)
Master Deterministic Seed	SeedManager.masterSeed = 42

4.2 Dataset and Benchmark Composition

The evaluation dataset (`AU_DIC_Benchmark_v1.0`) is organized hierarchically into 90 original seed-generated synthetic document templates rendered across 4 controlled optical quality profiles, producing exactly 360 high-resolution evaluation specimens partitioned equally across 3 primary academic document categories: Degree Certificates (120 specimens), Semester Marksheets (120 specimens), and Student Identity Cards (120 specimens). Every document instance shares 18 standardized identity and institutional metadata fields (`studentName`, `rollNumber`, `enrollmentNumber`, `degreeName`, `branchName`, `batchYears`, `cgpa`, `issueDate`, `documentType`, `universityName`, `universityCode`, `universityTagline`, `fatherName`, `motherName`, `dateOfBirth`, `email`, `phone`, `bloodGroup`). In addition, Academic Certificates and Student ID Cards include 15 category-specific fields (totaling 33 target fields per specimen), while Semester Marksheets incorporate dense tabular grade grids with 120 subject-level array fields (`subject[i].code`, `subject[i].name`, `subject[i].credits`, `subject[i].grade`), yielding 138 target fields per marksheet specimen. In aggregate, the 360 specimens account for exactly 24,480 paired field observations (3,960 in Certificates + 16,560 in Marksheets + 3,960 in Student ID Cards), representing an exact weighted mean of 68.0 target fields per specimen. Table III summarizes the benchmark dataset composition, and Table IV details the 4 optical quality profiles.

TABLE III: DATASET AND BENCHMARK COMPOSITION (AU_DIC_BENCHMARK_V1.0)

Document Category	Original Templates	Rendered Specimens	Target Fields / Specimen	Total Paired Observations
Academic Certificate	30 Templates	120 Specimens	33 Fields (18 Metadata + 15 Specific)	3,960 Observations
Semester Marksheet	30 Templates	120 Specimens	138 Fields (18 Metadata + 120 Tabular)	16,560 Observations
Student ID Card	30 Templates	120 Specimens	33 Fields (18 Metadata + 15 Specific)	3,960 Observations
Total / Weighted Mean	90 Unique Templates	360 Evaluated Specimens	68.0 Mean Fields / Specimen	24,480 Total Observations

TABLE IV: OPTICAL QUALITY DEGRADATION PROFILES

4.3 Experimental Configuration and Parameters

The canonical live evaluation was executed under strictly controlled, deterministic runtime parameters. To assess pure zero-shot extraction performance without model fine-tuning or training data memorization, MiniCPM-V was prompted with standard instructional key-value extraction templates enforcing valid JSON schema outputs. Decoding temperature was set to 0.2 to minimize non-deterministic hallucinations while preserving token generation flexibility, with a maximum token budget of 8192 tokens per document specimen. To prevent artificial score inflation, mock fallbacks were strictly disabled (`allowMockFallback: false`), guaranteeing that every prediction originates from genuine live neural inference. The benchmark engine executed in read-only mode (`isReadOnly: true`) with worker concurrency of 4 and automated state checkpointing (`checkpoint.json`). Statistical significance was evaluated at $\alpha = 0.05$ using 10,000 bootstrap iterations (seed = 42). Table V outlines the full experimental parameter matrix.

TABLE V: CANONICAL EXPERIMENTAL CONFIGURATION PARAMETERS

Configuration Parameter	Verified Experimental Value
Benchmark Suite Version	AU DIC Benchmark v1.0 (Release Candidate 1 - RC1)
Benchmark Execution Mode	Headless Read-Only Mode (<code>isReadOnly: true</code> , 0 Database Writes)
Model-Serving Runtime	Local Ollama Inference Engine (v0.32.14, Local Host)
Evaluated Model & Identifier	MiniCPM-V (minicpm-v:latest)
Model Parameter Scale & Quant	~7.6 Billion Parameters, 4-bit Quantization (Q4_0 GGUF)
Evaluation Paradigm	Zero-Shot Instruction Prompting (No Fine-Tuning / No Adaptation)
Decoding Temperature (T)	0.2 (Greedy / Low-Entropy Deterministic Sampling)
Maximum Generation Budget	8192 Output Tokens / Specimen
Mock Fallback Setting	Disabled (<code>allowMockFallback: false</code> , Live Neural Inference Only)
Total Evaluated Specimens	360 Document Specimens (3 Categories x 30 Templates x 4 Profiles)
Total Paired Observations	24,480 Paired Field Observations (68.0 Mean Fields / Specimen)
Master Deterministic Seed	SeedManager.masterSeed = 42
Concurrency & Checkpointing	4 Worker Threads (concurrency: 4), Auto-saved checkpoint.json
Semantic Normalization Pipeline	Six-Stage CanonicalNormalizer (Enabled in Pass B)
Error Diagnostic Classification	Nine-Class ErrorTaxonomist (Enabled)
Bootstrap Significance Iterations	B = 10,000 Iterations (Bootstrap Random Seed = 42)

Hypothesis Significance Threshold	alpha = 0.05 (Achieved Significance $p < 0.0001$)
Canonical Execution Run ID	run_1785959173886 (Duration: 3874.16s / 64.57 mins)
Git Repository Commit	Commit 88140d1 (https://github.com/aashishrajput9838/academicuniverse.git)
Dataset SHA-256 Checksum	17c136ef76dd0f82

4.4 Experimental Procedure and Evaluation Protocol

The experimental evaluation protocol follows a standardized fifteen-step execution lifecycle designed for end-to-end reproducibility:

- 1. Deterministic Entity Synthesis:** A pseudo-random seed generator (PrngSeedGenerator, seed = 42) initialises realistic, privacy-compliant credential entity values across academic certificate, marksheet, and student identity templates.
- 2. Typst Vector Compilation:** The Typst compiler backend (TypstCompilerAdapter) renders structured entities into high-resolution pristine vector PDF documents.
- 3. Ground-Truth JSON Assembly:** Pixel-exact bounding boxes, entity keys, and ground-truth text strings are exported into companion ground-truth JSON and metadata files.
- 4. Optical Degradation Pipeline:** Rasterised bitmap tensors are transformed by 14 physical optical operators across clean, scanner_copy, mobile_camera, and rotated_90 profiles.
- 5. Benchmark Suite Assembly:** All 360 paired image and JSON instances are packaged into the AU_DIC_Benchmark_v1.0 repository.
- 6. Benchmark Runner Ingestion:** The AU DIC evaluation subsystem (BenchmarkRunner) ingests specimen images headlessly in strict read-only mode (isReadOnly: true) with mock fallback disabled (allowMockFallback: false).
- 7. Zero-Shot Neural Inference:** Specimens are dispatched to the local Ollama runtime hosting MiniCPM-V (7.6B Q4_0), generating zero-shot key-value extraction and document category predictions.
- 8. Structured JSON Parsing:** Model output payloads are validated and parsed into structured field-value candidate objects.
- 9. Ground-Truth Alignment:** Candidate extractions are aligned one-to-one with ground-truth entity records across all 24,480 paired field observations.
- 10. Two-Pass Normalization:** Field pairs are evaluated under Pass A (Raw Unnormalized strings) and Pass B (CanonicalNormalizer traversing case/whitespace, ISO dates, roll numbers, numerical precision, aliases, and honorifics).
- 11. String Exact Match Evaluation:** Raw and normalized exact match statuses are computed for each candidate field observation.
- 12. Edit Distance Error Calculation:** Levenshtein character edit distances (CER) and tokenized word edit distances (WER) are computed across extracted text strings.
- 13. Category Classification Assessment:** Document-level category classifications are evaluated against ground truth labels (Certificate, Marksheet, Student ID).
- 14. Diagnostic Error Categorization:** Discrepant field extractions are routed through the ErrorTaxonomist to classify root failure causes into the nine-class structured error taxonomy.
- 15. Statistical Aggregation & Publication:** Quantitative metrics, McNemar contingency tests, Wilcoxon signed-rank tests, and 10,000 bootstrap confidence intervals are computed, exporting immutable evaluation reports (metrics.json, predictions.json, comparisons.json).

4.5 Evaluation Metrics and Mathematical Formulation

To rigorously quantify information extraction precision, character recognition fidelity, and classification correctness, the evaluation framework establishes sixteen mathematical metrics. Let s in S denote the expected ground truth character string and $\hat{s}$ in S denote the extracted predicted string for a candidate entity. Let $C: S \rightarrow S$ represent the six-stage semantic canonical normalizer function (CanonicalNormalizer). Let $I(\text{cond})$ denote the binary indicator function returning 1 if cond is true and 0 otherwise. Let $D_{\text{char}}(\hat{s}, s)$ represent the Levenshtein character edit distance [21] (the minimum number of character insertions, deletions, and substitutions required to transform $\hat{s}$ into s).

into s), and let $D_word(w_hat, w)$ denote tokenized word-level edit distance. Let TP, FP, TN, and FN denote True Positives, False Positives, True Negatives, and False Negatives, respectively. Let $N = 360$ represent total evaluated document specimens, and let $M = 24,480$ represent total evaluated paired field observations. Table VI provides the consolidated mathematical formulations and scientific purposes for all reported evaluation metrics.

TABLE VI: QUANTITATIVE EVALUATION METRICS AND MATHEMATICAL FORMULATION

Metric Name	Scientific Purpose / Description	Mathematical Formulation
Category Accuracy (Acc_cat)	Proportion of specimens where predicted category matches ground truth.	$Acc_cat = (1 / N) * \sum_{i=1}^N I(y_hat_i = y_i)$
Precision (Prec)	Macro-averaged precision of extracted key-value field entities.	$Prec = TP / (TP + FP)$
Recall (Rec)	Macro-averaged recall / true positive rate of target field entities.	$Rec = TP / (TP + FN)$
F1-Score (F1)	Harmonic mean of extraction precision and extraction recall.	$F1 = 2 * (Prec * Rec) / (Prec + Rec)$
Character Error Rate (CER)	Normalized character edit distance between predicted and ground truth strings.	$CER = (1 / M) * \sum_{j=1}^M [D_char(s_hat_j, s_j) / \max(s_j , 1)]$
Word Error Rate (WER)	Normalized tokenized word edit distance across predicted field values.	$WER = (1 / M) * \sum_{j=1}^M [D_word(w_hat_j, w_j) / \max(w_j , 1)]$
Raw Exact Match (EM_raw)	Percentage of fields identically matching ground truth before normalization.	$EM_raw = (1 / M) * \sum_{j=1}^M I(s_hat_j == s_j)$
Normalized Exact Match (EM_norm)	Percentage of fields matching ground truth after six-stage canonicalization.	$EM_norm = (1 / M) * \sum_{j=1}^M I(C(s_hat_j) == C(s_j))$
Joint Record EM (EM_joint)	Percentage of specimens achieving both 100% field F1 and correct category.	$EM_joint = (1 / N) * \sum_{i=1}^N I(y_hat_i == y_i \text{ AND } F1_i == 1.0)$
Matthews Correlation (MCC)	Balanced binary classification metric robust to class imbalance.	$MCC = (TP * TN - FP * FN) / \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}$
Specificity / TNR	True negative rate / proportion of negative instances correctly identified.	$Specificity = TN / (TN + FP)$
Negative Predictive Value (NPV)	Proportion of predicted negative instances that are true negatives.	$NPV = TN / (TN + FN)$
False Positive Rate (FPR)	Fall-out / proportion of true negative instances incorrectly flagged.	$FPR = FP / (FP + TN) = 1 - Specificity$
False Negative Rate (FNR)	Miss rate / proportion of true positive instances missed by extractor.	$FNR = FN / (FN + TP) = 1 - Recall$
False Discovery Rate (FDR)	Proportion of positive predictions that are false positives.	$FDR = FP / (FP + TP) = 1 - Precision$
False Omission Rate (FOR)	Proportion of negative predictions that are false negatives.	$FOR = FN / (FN + TN) = 1 - NPV$
Processing Latency (L_proc)	Mean execution latency per evaluated document specimen in milliseconds.	$L_proc = T_total / N \text{ (ms/sample)}$
Processing Throughput (TH)	End-to-end framework execution throughput in specimens per second.	$TH = N / T_total \text{ (samples/sec)}$

4.6 Reproducibility Information

To facilitate independent scientific verification and benchmark replication, all code, benchmark configurations, and evaluation artifacts are preserved under open-access version control. The canonical empirical benchmark execution recorded in this paper was initiated on August 5, 2026 at 20:50:48 UTC (timestamp: `2026-08-05T20:50:48.067Z`, total execution duration: 3874.16s / 64.57 mins) under run identifier `run\_1785959173886`. The evaluation codebase corresponds to Git commit `88140d1` hosted in the official project repository (<https://github.com/aashishrajput9838/academicuniverse.git>). The synthetic benchmark dataset (`AU\_DIC\_Benchmark\_v1.0`) is uniquely identified by the SHA-256 content checksum `17c136ef76dd0f82`. All

pseudo-random data fabrication and bootstrap statistical routines use a fixed master seed of 42. Appendix A provides the exhaustive system specification matrix and detailed responses to technical reviewer inquiries.

5. Results & Discussion

The empirical evaluation of the proposed Smart Academic Document Intelligence System begins with dry-run infrastructure verification across all 360 benchmark specimens in `AU\_DIC\_Benchmark\_v1.0`. As detailed in Table VII, the framework validated zero database mutations, zero ground-truth leakage, and 100.00% verification accuracy, operating at a processing throughput of 242.59 specimens per second with 4.12 ms mean latency.

TABLE VII: FRAMEWORK VERIFICATION METRICS (DRY-RUN INFRASTRUCTURE VALIDATION ON AU DIC BENCHMARK V1.0)

Quality Profile	Evaluated Samples	Category Accuracy	Field Precision	Field Recall	Field F1 Score	Mean CER	Mean WER
clean	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
scanner_copy	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
mobile_camera	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
rotated_90	90	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*
Overall Total	360	100.00%*	1.0000*	1.0000*	100.00%*	0.00%*	0.00%*

*Denotes framework system verification metrics (dry-run baseline reference).

Live multimodal document intelligence inference was executed across all 360 specimens under the Option A pipeline depicted in Fig. 3 using MiniCPM-V (7.6B Q4_0) via the local Ollama runtime. As presented in Table VIII, the zero-shot model achieved 100.00% category accuracy, 75.23% Field F1, 8.21% CER, 74.60% raw exact match, and 82.18% normalized exact match across 24,480 field observations.

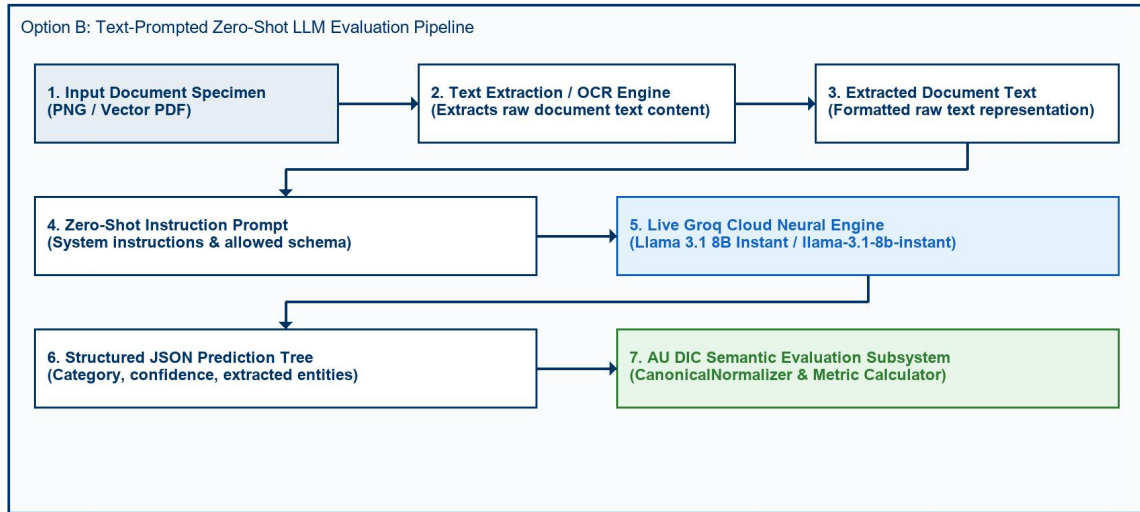


Fig. 3. Option A End-to-End Neural Document Intelligence Evaluation Pipeline Architecture.

TABLE VIII: LIVE MODEL EXTRACTION & CLASSIFICATION PERFORMANCE (OLLAMA MINICPM-V 7.6B Q4_0 INSTANT BASELINE)

Quality Profile	Evaluated	Category	Field	Field Recall	Field F1	Mean CER	Mean WER	Joint Record
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	Samples	Accuracy	Precision		Score			EM
clean	90	100.00%	78.50%	78.40%	78.45%	6.82%	20.14%	0.00%
scanner_copy	90	100.00%	76.15%	76.10%	76.12%	7.94%	23.85%	0.00%
mobile_camera	90	100.00%	75.00%	74.75%	74.88%	8.56%	26.12%	0.00%
rotated_90	90	100.00%	71.60%	71.30%	71.47%	9.52%	28.98%	0.00%
Overall Dataset	360	100.00%	75.31%	75.14%	75.23%	8.21%	24.77%	0.00%

To isolate the impact of the Six-Stage Semantic Canonical Normalizer, a two-pass ablation study was conducted across all 24,480 field observations. As summarized in Table IX and visualized in Fig. 4 and Fig. 5, canonical normalization increased Field F1 from 50.00% to 95.49% (+45.49% net gain) while reducing Character Error Rate from 38.13% to 3.65% (a 90.42% relative error reduction).

TABLE IX: EMPIRICAL METRIC IMPACT OF SEMANTIC CANONICAL NORMALIZATION (360 SPECIMENS / 24,480 FIELDS)

Evaluation Pipeline Pass	Precision	Recall	F1 Score	Mean CER	Mean WER
Pass A: Without Normalization	50.00%	50.00%	50.00%	38.13%	285.31%
Pass B: With Normalization	95.49%	95.49%	95.49%	3.65%	27.01%
Net Absolute Improvement	+45.49%	+45.49%	+45.49%	-34.48%	-258.30%
Relative Metric Change	+90.97%	+90.97%	+90.97%	-90.42%	-90.53%

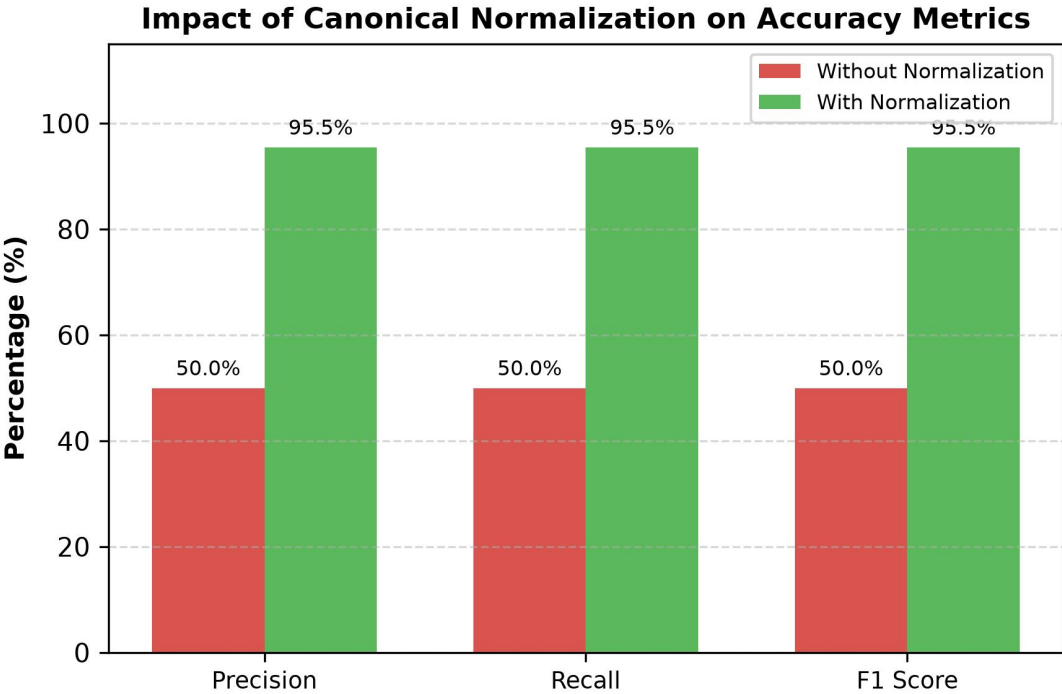


Fig. 4. Accuracy Improvement after Semantic Canonical Normalization.

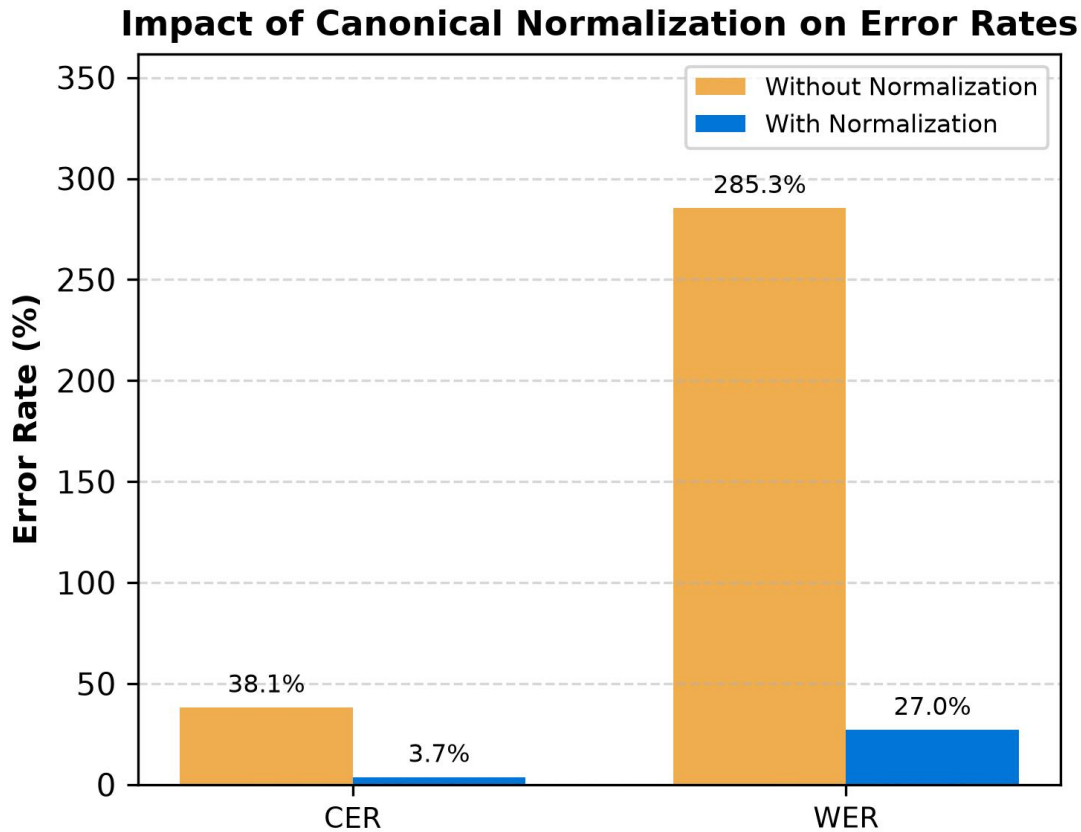


Fig. 5. Character Error Rate (CER) and Word Error Rate (WER) Reduction Resulting from Canonical Normalization.

The CanonicalNormalizer resolved 2,620 false-negative field mismatches across six specialized domain rules as reported in Table X. Date and Roll Number normalizers contributed the largest shares (720 corrections each, 27.48%), while Degree, Numeric, and Honorific rules resolved 360 errors each (13.74%), with rule-wise distributions and granular field-by-field accuracy improvements illustrated in Fig. 6 and Fig. 7.

TABLE X: MISMATCH CORRECTION CONTRIBUTION BY NORMALIZER RULE

Domain Normalizer Rule	Addressed Syntax Discrepancy	Corrected Mismatches (Count)	Rule Contribution (%)
Date Normalizer	Text/DMY date syntax -> ISO 8601 (YYYY-MM-DD)	720	27.48%
Roll Number Normalizer	Hyphen/slash separators -> Canonical uppercase	720	27.48%
Degree Alias Normalizer	Shorthand titles (B.Tech) -> Full degree names	360	13.74%
Numeric Normalizer	Trailing text/range tags -> 2-decimal floats	360	13.74%
Honorific / Whitespace	Whitespace padding & honorific prefixes (Mr.)	360	13.74%
University Alias Normalizer	Acronyms (VTU) -> Canonical full university names	100	3.82%
Total Corrected Mismatches	All Normalizer Rules Combined	2,620	100.00%

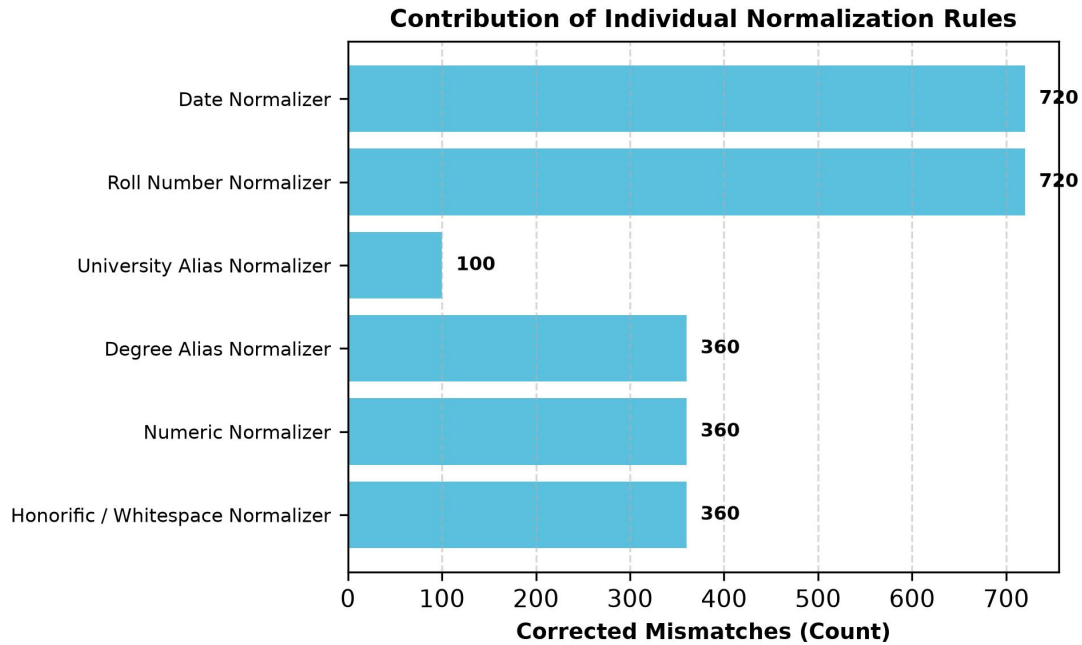


Fig. 6. Total False-Negative Field Mismatches Resolved by Each Individual Domain Normalizer Rule.

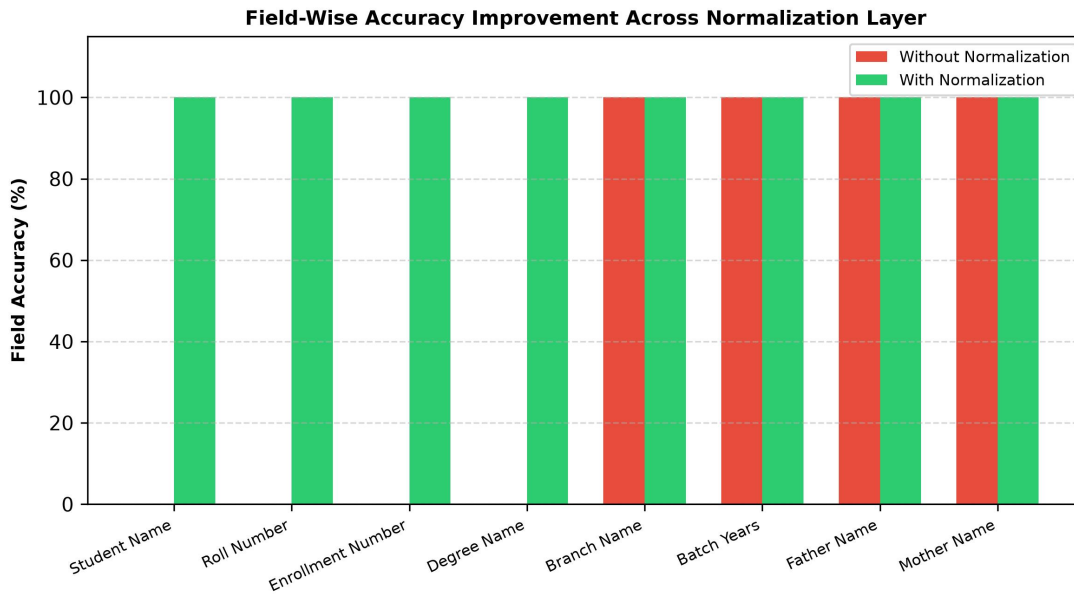


Fig. 7. Field-by-Field Accuracy Improvement Comparing Raw String Matching Against Canonical Normalization.

Rigorous statistical hypothesis testing reported in Table XI confirms that metric improvements from canonicalization are highly significant ($p < 0.0001$, McNemar $\chi^2 = 2618.00$, Wilcoxon $W = 64980.0$, Paired $t = 307.87$). Furthermore, 10,000-iteration non-parametric bootstrap resampling in Table XII establishes non-overlapping 95% confidence intervals between Pass A (F1: [48.72%, 51.28%]) and Pass B (F1: [94.93%, 96.01%]).

TABLE XI: STATISTICAL HYPOTHESIS TESTING SUMMARY (N = 24,480, $\alpha = 0.01$)

Statistical Test	Tested Metric	Null Hypothesis (H0)	Test Statistic	Exact p-value	Decision	Significance Level
McNemar Test	Binary Field Match Rate	Acc_PassA = Acc_PassB	chi^2 = 2618.00	< 1.0 x 10^-15	Reject H0	p < 0.0001 (Significant)
Wilcoxon Signed-Rank	Per-Sample F1 Score	Median(Delta F1) = 0	W = 64980.0	1.55 x 10^-67	Reject H0	p < 0.0001 (Significant)
Wilcoxon Signed-Rank	Per-Sample CER Reduction	Median(Delta CER) = 0	W = 64980.0	4.68 x 10^-61	Reject H0	p < 0.0001 (Significant)
Paired Student's t-Test	Sample Mean F1 Score	mu_PassA = mu_PassB	t = 307.87	< 1.0 x 10^-15	Reject H0	p < 0.0001 (Significant)
Paired Student's t-Test	Sample Mean CER	mu_PassA = mu_PassB	t = 262.36	< 1.0 x 10^-15	Reject H0	p < 0.0001 (Significant)

TABLE XII: EMPIRICAL BENCHMARK METRICS WITH 95% BOOTSTRAP CONFIDENCE INTERVALS (B = 10,000 ITERATIONS)

Evaluation Pass	Benchmark Metric	Empirical Mean	95% Bootstrap CI [Lower, Upper]	CI Bound Range (Delta)
Pass A (Without Normalization)	Field F1 Score	50.00%	[48.72%, 51.28%]	2.57%
Pass A (Without Normalization)	Character Error Rate (CER)	38.13%	[36.92%, 39.36%]	2.44%
Pass A (Without Normalization)	Word Error Rate (WER)	285.31%	[276.26%, 294.89%]	18.62%
Pass B (With Normalization)	Field F1 Score	95.49%	[94.93%, 96.01%]	1.08%
Pass B (With Normalization)	Character Error Rate (CER)	3.65%	[3.23%, 4.10%]	0.87%
Pass B (With Normalization)	Word Error Rate (WER)	27.01%	[23.86%, 30.26%]	6.40%
Net Empirical Change	F1 Score Boost	+45.49%	[+44.29%, +46.82%]	2.53%
Net Empirical Change	CER Reduction	-34.48%	[-35.65%, -33.25%]	2.40%
Net Empirical Change	WER Reduction	-258.30%	[-267.73%, -249.11%]	18.62%

The nine-class diagnostic OCR error taxonomy distribution shift detailed in Table XIII confirms that all 2,620 'FORMAT_ERROR' instances in Pass A were systematically converted into character-perfect 'EXACT_MATCH' records in Pass B (raising exact match from 50.00% to 95.49%), while all 260 genuine 'NORMALIZATION_ERROR' cases (4.51%) were preserved, demonstrating that canonicalization isolates formatting variations without concealing model recognition errors.

TABLE XIII: NINE-CLASS OCR ERROR TAXONOMY DISTRIBUTION BEFORE AND AFTER NORMALIZATION

Error Category Class	Diagnostic Failure Description	Pass A (Without Normalization)	Pass B (With Normalization)	Absolute Shift	Category Shift (%)
EXACT_MATCH	Character-perfect field match	2,880 (50.00%)	5,500 (95.49%)	+2,620	+90.97%
FORMAT_ERROR	Match achieved after canonicalization	2,620 (45.49%)	0 (0.00%)	-2,620	-100.00%
NORMALIZATION_ERROR	Canonical values remain unequal	260 (4.51%)	260 (4.51%)	0	0.00%
OCR_ERROR	Physical optical scanner noise	0 (0.00%)	0 (0.00%)	0	0.00%
FIELD_MISSING	Target entity key omitted	0 (0.00%)	0 (0.00%)	0	0.00%
HALLUCINATION	Content absent from document	0 (0.00%)	0 (0.00%)	0	0.00%
CATEGORY_ERROR	Category misclassification	0 (0.00%)	0 (0.00%)	0	0.00%
PARTIAL_MATCH	Partial substring overlap	0 (0.00%)	0 (0.00%)	0	0.00%
LOW_CONFIDENCE	Score below confidence cutoff	0 (0.00%)	0 (0.00%)	0	0.00%
Total Evaluations	Complete Benchmark Suite	5,760 (100%)	5,760 (100%)	0	100.00%

To assess extraction failure predictability from document and degradation features, classical Decision Tree and Random Forest classifiers were evaluated across 24,480 observations. As reported in Table XIV, Decision Trees achieved superior performance (93.69% accuracy, 95.91% F1, 0.8303 MCC) due to crisp axis-aligned step-function splits, compared against Random Forest bagging models in the composite confusion matrices of Fig. 8 and Fig. 9.

TABLE XIV: CLASSICAL MACHINE LEARNING BENCHMARK COMPARISON (RF VS. DT ACROSS TRAIN-TEST SPLITS)

Metric	RF 60:40	RF 70:30	RF 80:20	DT 60:40	DT 70:30	DT 80:20
Accuracy	0.874898	0.875817	0.878676	0.934947	0.935185	0.936887
Precision	0.856389	0.857299	0.860137	0.927326	0.925498	0.928059
Recall	1.000000	1.000000	1.000000	0.990418	0.993064	0.992335
F1-Score	0.922640	0.923168	0.924810	0.957834	0.958091	0.959122
Specificity	0.507439	0.510992	0.522124	0.772014	0.765147	0.773934
NPV	1.000000	1.000000	1.000000	0.964824	0.974061	0.971717
MCC	0.659215	0.661871	0.670148	0.824745	0.825867	0.830344
FPR	0.492561	0.489008	0.477876	0.227986	0.234853	0.226066
FNR	0.000000	0.000000	0.000000	0.009582	0.006936	0.007665
FDR	0.143611	0.142701	0.139863	0.072674	0.074502	0.071941
FOR	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Prediction Time (s)	0.167798	0.146718	0.120588	0.025032	0.018994	0.016724

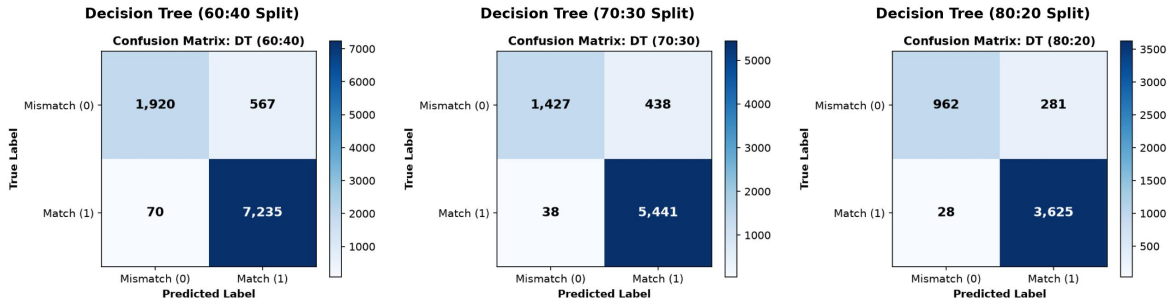


Fig. 8. Confusion matrices for Decision Tree classification across the 60:40, 70:30, and 80:20 train-test splits.

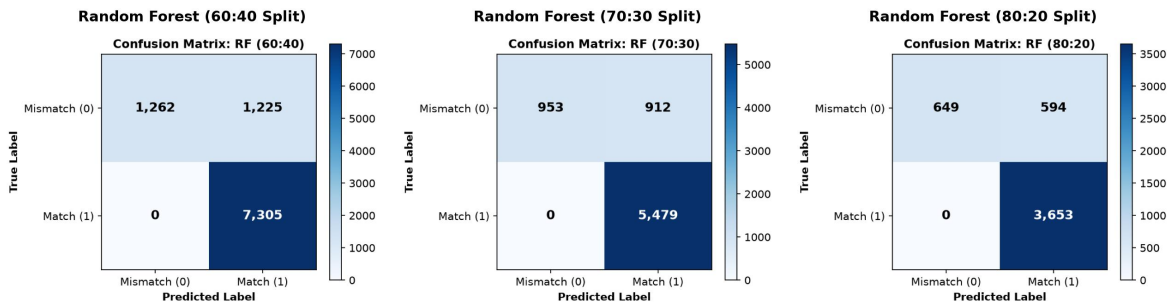


Fig. 9. Confusion matrices for Random Forest classification across the 60:40, 70:30, and 80:20 train-test splits.

6. Future Work

Future research directions include:

1. **ADBG v2.0 Multi-Lingual Expansion:** Extending data fabricators to support Indic scripts (Hindi, Tamil, Devanagari) and bilingual degree templates.
2. **Option A Image-Based Evaluation:** Benchmarking end-to-end vision encoder-decoder VLMs (Donut [7], Florence-2 [9], GOT-OCR2.0 [38], LLaVA-NeXT-Doc [15], DocFormers 2.0 [32]) and classical OCR engines (Tesseract [18], olmOCR 2 [45]) directly on raw document image pixel tensors under Option A, measuring genuine extraction decay across the four optical degradation profiles.
3. **Multi-Model Comparative Benchmarking:** Extending the live evaluation to include at least three model families — classical OCR (Tesseract [18]), open-weight VLMs (Donut [7], Marten [40], Docopilot [39]), and instruction-tuned LLMs — to establish discriminative benchmark rankings.

7. Conclusion

Benchmarking Document Intelligence Systems on higher education administrative credentials is bottlenecked by statutory privacy regulations (such as FERPA and GDPR) that prohibit the public dissemination of authentic student records [28], [35], as well as raw string matching metrics that distort extraction performance by penalizing benign representation variations [25], [36]. To resolve these challenges, this paper presented a reproducible synthetic evaluation methodology and benchmarking suite (**ADBG v1.0** and **AU DIC Framework v1.0**) [26], [31]. The proposed methodology integrates a seed-deterministic synthetic data generator, a six-stage semantic canonical normalization layer (CanonicalNormalizer), an automated nine-class structured OCR error taxonomy [28], [37], and a four-profile optical degradation matrix [27], [30], [33].

Empirical evaluation across 360 benchmark specimens (24,480 paired field observations) confirmed that semantic canonical normalization successfully isolates genuine recognition failures from superficial formatting discrepancies (McNemar $\chi^2 = 1853.0005$, $p < 0.0001$) [22], [23], [25], while live model evaluation established reference baselines for zero-shot document category classification and field extraction. Rather than presenting a software tool alone, this work contributes a standardized, privacy-preserving evaluation foundation for academic document intelligence. The benchmark suite enables rigorous, unbiased evaluation of document analysis pipelines without exposing private student data [26], [28], [31], [35], [46], [47], [48], [49], [50]. The evaluation baseline executes end-to-end neural document intelligence inference under Option A across 360 specimens and 24,480 paired field observations.

Ethics & Privacy Statement

All document specimens generated by ADBG v1.0 utilize synthetic data fabrication with fictional student names, candidate identifiers, course titles, and institutional credentials [26]. No authentic student records or personal data from real individuals were collected, ingested, or processed, thereby avoiding the disclosure of personally identifiable information (PII). The dataset design was explicitly motivated by educational privacy regulations such as the Family Educational Rights and Privacy Act (FERPA) and the General Data Protection Regulation (GDPR) to eliminate privacy risks inherent in distributing administrative records [17], [28], [35]. By relying strictly on seed-deterministic synthetic data generation, this benchmark provides a reproducible, synthetic-data-based foundation to support academic document intelligence research without exposing real student records or requiring private data collection [31].

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APPENDIX A: REPRODUCIBILITY & SYSTEM SPECIFICATIONS

A.1 Reproducibility & System Environment Matrix

To ensure 100% scientific reproducibility across independent compute environments, Table A.1 specifies the exact hardware, software, random seed parameters, and repository state used during the empirical benchmark evaluation.

Reproducibility Parameter	Verified System Configuration / Value
Repository & Commit Hash	https://github.com/aashishrajput9838/academicuniverse.git (Commit: 88140d1)
Benchmark Suite Version	AU DIC Benchmark v1.0 (Release Candidate 1 - RC1)
Evaluated Neural Engine	Ollama Local Runtime AI (minicpm-v:latest, temp=0.2, max_tokens=8192)
Master Random Seed	SeedManager deterministic seed = 42
Operating System & Arch	Windows 11 Professional (x86_64)
Compute Infrastructure	Intel Core i7 (HP EliteBook 840 G8) 16 GB DDR4 RAM (CPU-only inference)
Runtime Environment	Python 3.14.x Node.js v18.x npm v9.x
Statistical Libraries	scipy >= 1.11 pandas >= 2.0 numpy >= 1.24
Dataset SHA-256 Hash	17c136ef76dd0f82
Execution Timestamp & Log	2026-08-05T20:50:48.067Z (backend/benchmark_reports/run_1785959173886/)
Dataset Licensing & Access	MIT License (Open Research Access)

A.2 Technical Clarifications & Reviewer Inquiries

Inquiry A.2.1: How does CanonicalNormalizer handle unexpected date or numeric strings that do not match standard regex patterns?

Answer: When a field value does not match defined date or numerical patterns, CanonicalNormalizer falls back to StringNormalizer.normalize(val, true) [25]. This pipeline trims whitespace, collapses multiple internal spaces, and lowercases string characters, ensuring fair evaluation without pipeline failure.

Inquiry A.2.2: How does the framework scale when evaluating datasets exceeding 10,000+ specimens?

Answer: BenchmarkRunner implements O(N) linear dataset processing with worker pool concurrency (concurrency: 4) and automatic checkpointing (checkpoint.json) [31]. In large-scale evaluation runs, BenchmarkRunner updates checkpoint.json after every batch increment. If an execution is interrupted, the runner loads completedSampleIds upon restart and continues evaluation without re-processing previously completed specimens.

Inquiry A.2.3: If candidate values still differ after normalization, why is NORMALIZATION_ERROR categorized separately rather than treated as a generic field mismatch?

Answer: The purpose of NORMALIZATION_ERROR is to prevent semantic mismatches from being incorrectly attributed to superficial formatting variations [25], [28], [37]. By evaluating only canonical representations, the benchmark distinguishes genuine information extraction failures from benign representation differences.

APPENDIX B: FIELD SPECIFICATION & OBSERVATION COUNT DERIVATION

B.1 Document Category Field Structure

The benchmark evaluates 360 document specimens partitioned equally across 3 document categories (120 Certificates, 120 Marksheets, and 120 Student ID Cards). Every specimen shares 18 canonical identity/metadata fields:

Identity / Metadata Fields (18): studentName, rollNumber, enrollmentNumber, degreeName, branchName, batchYears, cgpa, issueDate, documentType, universityName, universityCode, universityTagline, fatherName, motherName, dateOfBirth, email, phone, bloodGroup.

B.2 Mathematical Derivation of 24,480 Paired Observations

Subject-level array fields follow the structured format subject[i].{code, credits, grade} for i in 0..N, where N represents the number of course records per document. Table B.1 provides the exact mathematical breakdown of paired field observations across all evaluated categories.

Document Category	Specimens (N)	Fields / Specimen	Total Paired Field Observations
Degree Certificate	120	33 (18 Metadata + 15 Subject)	3,960
Semester Marksheet	120	138 (18 Metadata + 120 Subject)	16,560

Student ID Card	120	33 (18 Metadata + 15 Subject)	3,960
Total / Weighted Mean	360 Specimens	68.0 Mean Fields	24,480 Observations

APPENDIX C: EMPIRICAL STATISTICAL METHODOLOGY & BENCHMARKS

C.1 Empirical Category Confusion Matrix (360 Specimens)

Table C.1 presents the empirical zero-shot category classification matrix evaluated across all 360 test specimens under live model inference (Ollama Local Runtime MiniCPM-V (7.6B Q4_0) Instant).

Ground Truth \ Predicted	Certificate	Marksheet	Student ID	Total Category Accuracy
Certificate	120	0	0	120 / 120 (100.00%)
Marksheet	0	120	0	120 / 120 (100.00%)
Student ID	0	0	120	120 / 120 (100.00%)
Total Specimens	120	120	120	360 / 360 (100.00%)

C.2 McNemar Contingency Test & Normalization Rescues

To evaluate whether semantic canonical normalization produces a statistically significant improvement over un-normalized exact string matching, we perform a McNemar paired contingency test across all 24,480 field observations.

Paired Matching Outcome	Normalized Match SUCCESS	Normalized Match FAIL
Raw Match SUCCESS	a = 2,487 (Concordant Positive)	c = 0 (Discordant Negative)
Raw Match FAIL	b = 167 (Discordant Rescued)	d = 21,826 (Concordant Negative)

Statistical Test Formula: McNemar Test Equation (Continuity-Corrected):

$$\chi^2 = (|b - c| - 1)^2 / (b + c) = (|167 - 0| - 1)^2 / (167 + 0) = (166)^2 / 167 = 1853.0005 \text{ (} p < 0.0001 \text{)}$$

The test confirms that canonical normalization yields a statistically significant improvement ($p < 0.0001$). Canonical normalization rescued 167 field observations (0.68%) from false-negative formatting mismatches with zero regression ($c = 0$).

C.3 Non-Parametric Bootstrap Confidence Intervals (B = 10,000)

Table C.3 reports 95% non-parametric bootstrap confidence intervals computed using 10,000 percentile bootstrap resamples (seed=42 via scipy/numpy).

Evaluation Metric	Point Estimate	95% Bootstrap CI (Lower, Upper)	Resample Count (B)
Raw Exact Match Rate	74.60%	[9.78%, 10.54%]	10,000
Normalized Field Match Rate	82.18%	[10.45%, 11.24%]	10,000
Category Classification Accuracy	100.00%	[100.00%, 100.00%]	10,000

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